

which refers to the acceleration of the electrons striking the metal target.

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Option D is true. Conservation of momentum: Total Initial momentum = 0.
Hence, when γ -photon emitted in one direction, the nucleus must recoil with an equal magnitude momentum in the opposite direction such that the total momentum remains 0.

Option A is not true. The greater the binding energy *per nucleon*, then the more stable the nucleus.

Option B is not true. The reaction is still possible with the provision of input of energy to the reactants.

Option C is not true. The half-life of a radioactive substance cannot be changed chemically.

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